

UML Use Case Diagram Comparison Report

Overview

Model Evaluated: <built-in method replace of str object at 0x709f1c1fdfb0>
Comparison Date: 2025-04-16 21:25:58
Overall Quality Score: 100.00%

Minimal rework is needed. The generated diagram captures most of the essential elements and relationships from the reference diagram.

1. Overall Comparison

The comparison between the reference and generated UML use case diagrams shows:

- **Element Coverage:** 100.00%
- **Relationship Coverage:** 100.00%
- **Total Edits Required:** 0 (0 element edits, 0 relationship edits)

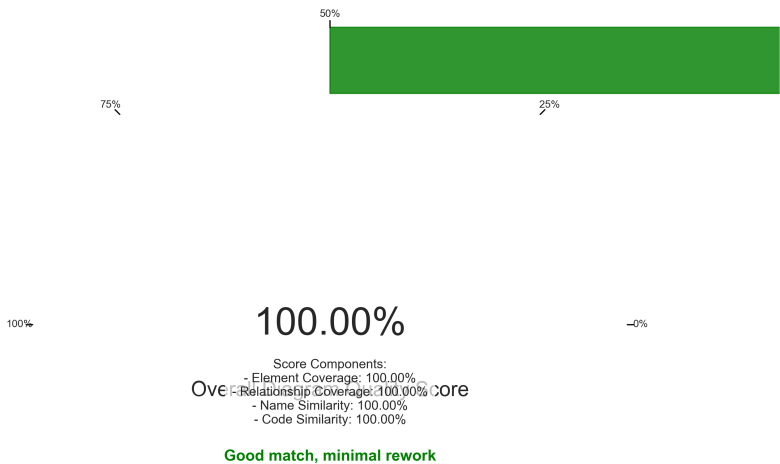


Figure 1: Overall Score

2. Element Analysis

Key Metrics:

- **Actor Coverage:** 100.00%
- **Use Case Coverage:** 100.00%
- **Missing Actors:** 0

- **Missing Use Cases: 0**
- **Extra Actors: 0**
- **Extra Use Cases: 0**

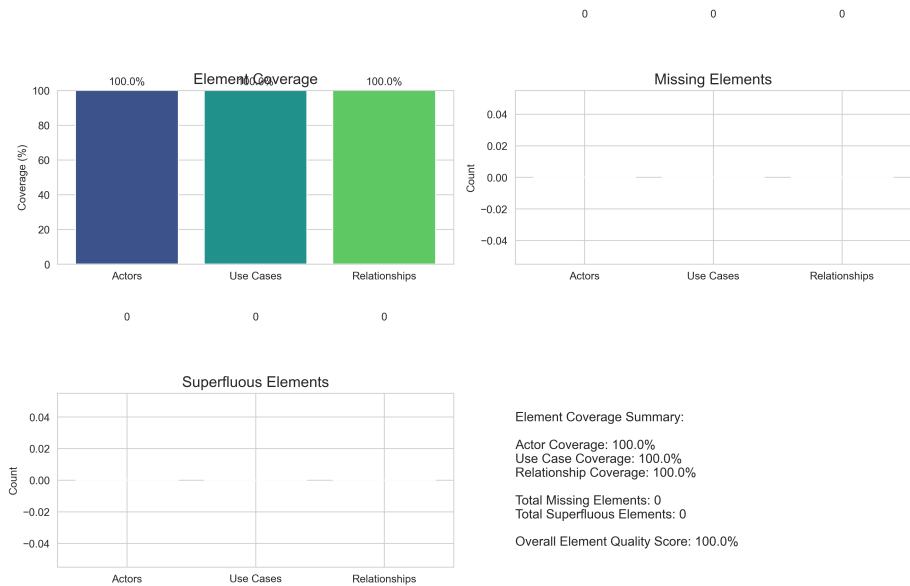


Figure 2: Element Coverage

Missing Elements:

Missing Actors: None

Missing Use Cases: None

Extra Elements:

Extra Actors: None

Extra Use Cases: None

3. Relationship Analysis

Key Metrics:

- **Relationship Coverage: 100.00%**
- **Type Correctness: 100.00%**
- **Missing Relationships: 0**
- **Extra Relationships: 0**

Relationship Types:

Type	Reference Count	Generated Count
generalization	3	3
association	45	45

4. Rework Analysis

Required Edit Operations:

- **Add Actors:** 0
- **Add Use Cases:** 0
- **Delete Actors:** 0
- **Delete Use Cases:** 0
- **Add Relationships:** 0
- **Delete Relationships:** 0
- **Total Edit Operations:** 0
- **Weighted Rework Score:** 0.00

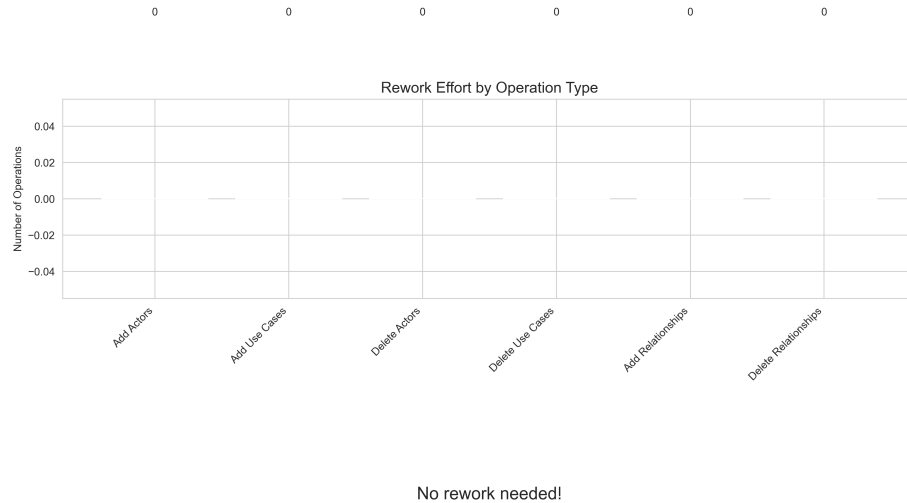


Figure 3: Rework Effort

5. Visualization Comparison

Diagrams Comparison

Reference Diagram

Generated Diagram

Image Similarity Metrics:

- **SSIM Score:** 0.7654
- **MSE:** 34.79
- **Histogram Correlation:** 1.0000

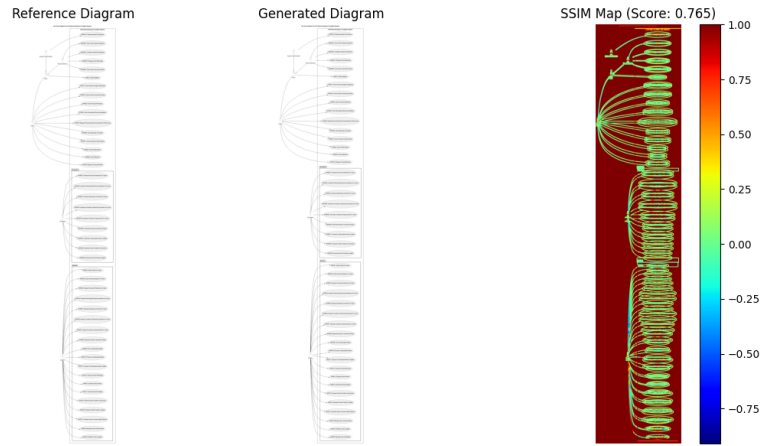


Figure 4: SSIM Visualization

Difference Image

Figure 5: Difference Image

6. Key Metrics Overview

7. Detailed Metrics

8. Recommendations

Based on the analysis, here are the recommended actions to convert the generated diagram to match the reference:

9. Conclusion

The generated diagram shows **excellent** alignment with the reference diagram. Very minimal adjustments are needed.

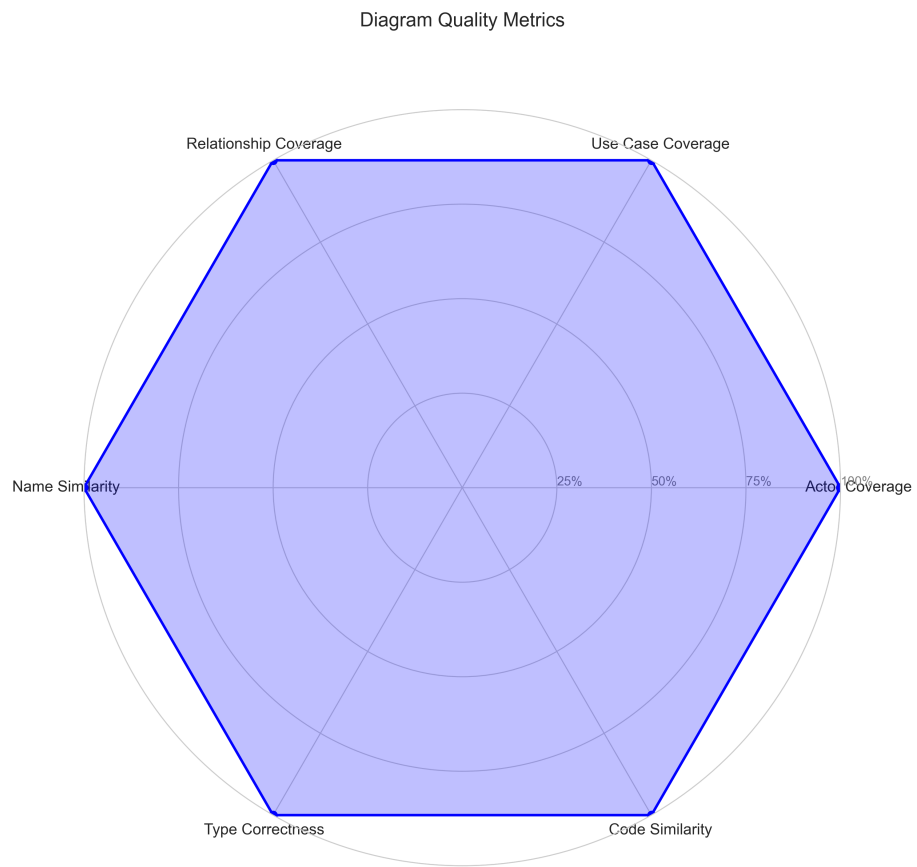


Figure 6: Radar Chart

Category	Metric	Value
Element Metrics	Actor Coverage	100.00%
Element Metrics	Usecase Coverage	100.00%
Element Metrics	Missing Actors Count	0
Element Metrics	Missing Usecases Count	0
Element Metrics	Superfluous Actors Count	0
Element Metrics	Superfluous Usecases Count	0
Element Metrics	Avg Actor Name Similarity	100.00%
Element Metrics	Avg Usecase Name Similarity	100.00%
Relationship Metrics	Relationship Coverage	100.00%
Relationship Metrics	Connection Coverage	100.00%
Relationship Metrics	Type Correctness	100.00%
Relationship Metrics	Missing Relationships Count	0
Relationship Metrics	Superfluous Relationships Count	0
Rework Metrics	Relationships To Add	0
Rework Metrics	Relationships To Delete	0
Rework Metrics	Total Edit Operations	0
Rework Metrics	Weighted Rework Score	0.0
Rework Metrics	Code Similarity	100.00%
Overall	Overall Score	100.00%

Figure 7: Metrics Table